

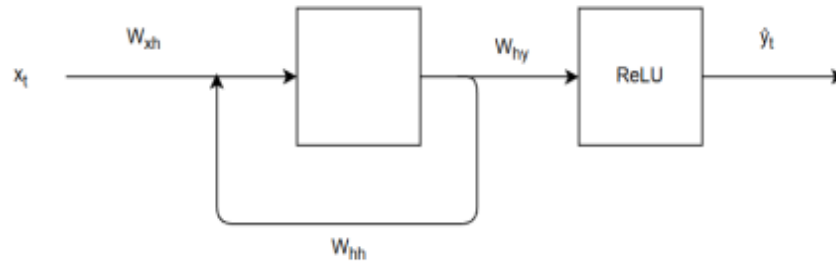
Brac University
CSE427: Machine Learning
Summer 2026
Quiz 05
Set A

Duration: 35 minutes

Name:	ID:	Section:
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Question 1:

[4]



$$\text{Loss (SSR)} = \sum_{i=1}^t (y_i - \hat{y}_i)^2$$

Assume, you have five data points of a sequence and you are trying to train the above RNN architecture.

Find the expression for $\frac{d\text{Loss}}{dW_{xh}}$ when the 5th datapoint (x_5) will be passed.

Question 2:

[4+2]

A CNN is applied to a RGB image of size 32×32 and has the following layers, in order.

Layer 1: CONV 3×3 , stride 1, padding 2, 16 filters

Layer 2: CONV 3×3 , stride 1, padding 0, 32 filters

Layer 3: POOL 2×2 , stride 2

- Find** the output size at each layer and total number of parameters in the whole network.
- What** will be the dimensions of the image matrix, the filter matrix, and the resulting output matrix of the convolution at Layer 2.

Question - 1

$$\frac{dLoss}{dw_{xh}} = \frac{dLoss}{d\hat{y}_5} \times \frac{d\hat{y}_5}{dw_{xh}}$$

$$Loss (SSR) = \sum_{i=1}^t (y_i - \hat{y}_i)^2$$

$$\begin{aligned} \frac{dLoss}{d\hat{y}_5} &= \frac{d}{d\hat{y}_5} \left[\sum_{i=1}^t (y_i - \hat{y}_i)^2 \right] \\ &= -2 \sum_{i=1}^t (y_i - \hat{y}_i) \quad \text{or} \quad -2 (y_5 - \hat{y}_5) \\ &\quad \left[\text{for all other } y \text{ the differentiation} = 0 \right] \end{aligned}$$

$$\hat{y}_5 = \text{ReLU}(z_5)$$

$$\begin{aligned} z_5 &= (((x_1 w_{xh} w_{hh} + x_2 w_{xh}) w_{hh} + x_3 w_{xh}) w_{hh} + x_4 w_{xh}) \\ &\quad w_{hh} + x_5 w_{xh}) w_{hy} \\ &= (x_1 w_{xh} w_{hh}^4 + x_2 w_{xh} w_{hh}^3 + x_3 w_{xh} w_{hh}^2 + x_4 w_{xh} w_{hh} \\ &\quad + x_5 w_{xh}) w_{hy} \end{aligned}$$

$$\text{Now, } \frac{d\hat{y}_5}{dw_{xh}} = \frac{d\hat{y}_5}{dz_5} \times \frac{dz_5}{dw_{xh}} \quad \left| \begin{array}{l} \frac{d\hat{y}_5}{dz_5} \\ \frac{dz_5}{dw_{xh}} \end{array} \right.$$

$$\frac{d\hat{y}_5}{dz_5} = \frac{d}{dz_5} \text{ReLU}(z_5) = \text{ReLU}'(z_5)$$

$$\begin{aligned} \frac{dz_5}{dw_{xh}} &= \frac{d}{dw_{xh}} (x_1 w_{xh} w_{hh}^4 + x_2 w_{xh} w_{hh}^3 + x_3 w_{xh} w_{hh}^2 + \\ &\quad x_4 w_{xh} w_{hh} + x_5 w_{xh}) w_{hy} \\ &= (x_1 w_{hh}^4 + x_2 w_{hh}^3 + x_3 w_{hh}^2 + x_4 w_{hh} + x_5) w_{hy} \end{aligned}$$

$$\therefore \frac{d\text{Loss}}{dW_{nn}} = -2 \sum_{i=1}^t (y_i - \hat{y}_i) \times \text{ReLU}'(z_5) \times (x_1 W_{nn}^4 + x_2 W_{nn}^3 + x_3 W_{nn}^2 + x_4 W_{nn} + x_5) w_{ny}$$

Question-2

[a] Given, RGB image of size 32×32
 so, image dimension $32 \times 32 \times 3$ [RGB $\rightarrow$ Channel = 3
 Grayscale $\rightarrow$ " = 1]
 so, $W = H = 32$

Layer-1 conv 3×3 , stride, $s=1$, padding, $P=2$

No. of filter = 16

$$\text{Output} = \frac{W - F + 2P}{s} + 1 = \frac{32 - 3 + 2 \times 2}{1} + 1 = 34$$

so, Feature map dimension $34 \times 34 \times 16$
 (as $W = H = 32$) $\rightarrow$ No. of filter

Parameters = Dimension of filter $\times$ No. of filter +
 No. of filter

$$= (3 \times 3 \times 3) \times 16 + 16$$

$$= 448$$

[Conv 3×3 , image channel = 3,
 so, filter channel = 3]

Layer - 2 conv - 3×3 , stride 1, padding 0, 32 filters.

$$\text{Output size} = \frac{34 - 3 + 2 \times 0}{1} + 1 \quad \left[\begin{array}{l} \text{Last layer output} \\ 34 \times 34 \times 16 \\ W=H=34 \end{array} \right]$$
$$= 32$$

So, Feature Map dimension $32 \times 32 \times 32$ No. of filter

$$\text{Parameters} = (3 \times 3 \times 16) \times 32 + 32 = 4640$$

$\downarrow \qquad \qquad \downarrow$
[conv - 3×3] [image/output
was $34 \times 34 \times 16$
So, Filter channel = 16]

Layer - 3 POOL 2×2 , stride 1, $s=2$ [there is no padding in Pooling Layer]

$$\text{Output size} = \frac{W - F}{s} + 1$$
$$= \frac{32 - 2}{2} + 1$$
$$= 16$$

[previous layer
output $32 \times 32 \times 32$
 $W=H=32$]

So, Feature Map dimension $16 \times 16 \times 32$

Parameters = 0 [Pooling Layer has no Learnable weight]

$$\text{So, Total Parameters} = 448 + 4640 + 0 = 5088$$

[6] At Layer-2,

$$\text{input dimension } 34 \times 34 \times 16$$

$$\text{Filter dimension } 3 \times 3 \times 16$$

$$\text{Output Feature Map dimension } 32 \times 32 \times 32$$

$$\text{No. of filters} = 32$$

Image Matrix

$$\text{output dimension } 32 \times 32 \times 32 \rightarrow \text{channel}$$

$$\text{So, No. of blocks in image matrix}$$

$$= 32 \times 32 \quad [\text{blocks in each channel}]$$

$$= 1024$$

$$\text{Filter dimension} = 3 \times 3 \times 16$$

$$\text{So, No. of pixels in each block} = 3 \times 3 \times 16$$

$$= 144$$

$$\text{So, Image Matrix size} = 144 \times 1024$$

Filter Matrix

$$\text{No. of pixels in Each Filter} = 3 \times 3 \times 16 \\ = 144$$

[same as No. of
pixels in each
block]

$$\text{No. of Filters in Layer-2} = 32$$

$$\text{So, Filter Matrix Size} = 32 \times 144$$

Output Matrix

$$\text{output Matrix} = (\text{Filter Matrix}) \times (\text{Input Matrix})$$

$$= (32 \times 144) \times (144 \times 1024)$$

$$= 32 \times 1024$$

$$= 32 \times \underline{32 \times 32} \quad [\text{Reshaped}]$$



dimension

No. of
Channel